Discrete Electronics Project: Class AB Audio Amplifier
Design and Simulation Report

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***Abstract*—This paper presents the design and simulation of a discrete-component Class AB audio amplifier developed by Analog One. The system combines signal conditioning, voltage amplification, and a power stage to drive an $8\,\Omega$ loudspeaker. The design uses commercial component values and considers signal quality, transistor bias, and heat dissipation. The main calculations establish a $20\,\mathrm{W}$ output reference. In the reported power-stage test, a $20\,\mathrm{V}$ peak input at $10\,\mathrm{kHz}$ produces $22.8\,\mathrm{W}$ of average load power. Calculated requirements and simulation readings are presented in separate tables to support a clear assessment of the design.**

***Index Terms*—Class AB, audio amplifier, discrete circuits, MOSFET, circuit simulation.**

I. Project Description

The project develops an audio amplifier that increases the level of an incoming audio signal and supplies the power needed by an $8\,\Omega$ loudspeaker. Developed by Analog One, the design aims to reproduce the signal at a useful output level while limiting distortion and power losses.

The amplifier contains three stages. The preamplifier conditions the incoming signal, the voltage amplifier increases its amplitude, and the power stage provides the current required by the loudspeaker. The Class AB output arrangement uses two complementary transistors that share the audio waveform. A small resting current helps smooth the transition between the positive and negative portions of the signal.

The design combines calculations, commercial component values, and circuit simulation. The evaluation considers output power, signal transfer, bias adjustment, and heat dissipation, together with the interaction between the stages.

II. Block Diagrams

A. Overall Amplifier

Figure 1 shows how the signal moves through the amplifier. Each stage has a specific role, from preparing the incoming signal to delivering power to the loudspeaker.

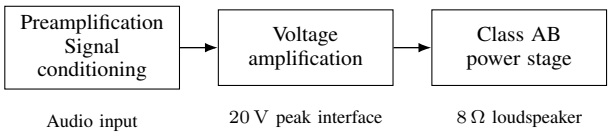


Figure 1. Main stages of the audio amplifier.

B. Preamplification Stage

C. Voltage Amplification Stage

D. Class AB Power Stage

The power stage uses an IRFZ44N and an IRF4905 MOSFET. These are complementary transistors: one provides most of the current during the positive portion of the signal, while the other handles most of the negative portion. Class AB bias keeps both slightly conducting around the transition, helping reduce crossover distortion [**?**, sec. 7–2].

The stage mainly increases the available current. Its output voltage follows the input with the same polarity and a small reduction in amplitude, as expected from a source-follower arrangement [**?**, sec. 9–2].

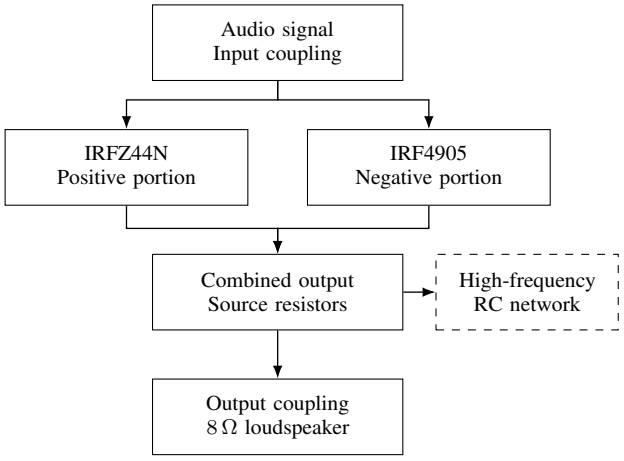


Figure 2. Functional blocks of the Class AB power stage. The RC network combines a resistor and a capacitor.

III. Design Considerations and Calculated Values

A. Preamplification Stage

B. Voltage Amplification Stage

C. Class AB Power Stage

Figure **??** presents the power-stage circuit. The design retains the complementary output arrangement, with nominal supplies of $+24\,\mathrm{V}$ and $-24\,\mathrm{V}$.

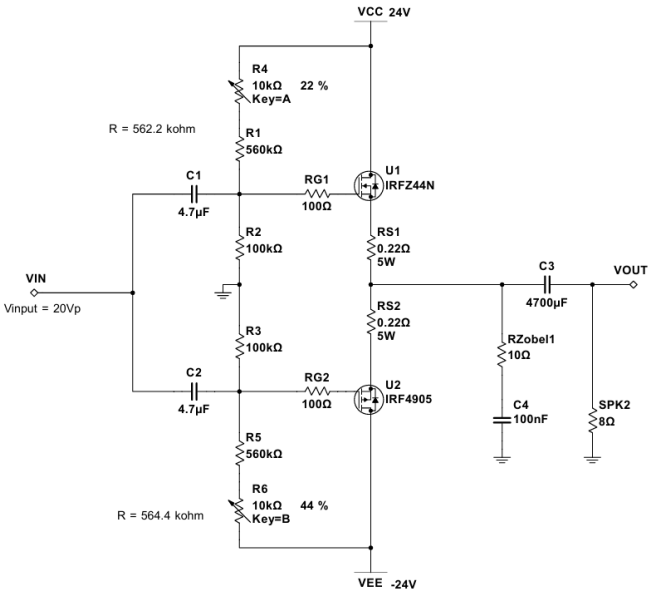


Figure 3. Class AB power-stage schematic with IRFZ44N and IRF4905 MOSFETs.

1) Output Power Requirement: The design reference is $20\,\mathrm{W}$ delivered to an $8\,\Omega$ load. For a sinusoidal signal, the average power is

$$P_L = \frac{V_{L,\mathrm{rms}}^2}{R_L}. \tag{1}$$

Here, RMS is the effective voltage value used to calculate average power of a signal audio. For this project the required voltage and current of this signal is:

$$V_{LOAD,\mathrm{rms}} = \sqrt{(20)(8)} \approx 12.65\,\mathrm{V}, \tag{2}$$

$$V_{LOAD,peak} = \sqrt{2}(12.65) \approx 17.89\,\mathrm{V}, \tag{3}$$

$$I_{LOAD,\mathrm{rms}} = \frac{12.65}{8} \approx 1.58\,\mathrm{A}. \tag{4}$$

These values define the load requirements. With the specified $20\,\mathrm{V}$ peak input, the output power depends on the voltage transfer of the stage. Table **??** summarizes the calculated values.

Table I
Calculated Output Requirements for the 20 W Reference

Quantity	Calculated value
Average output power, reference	$20\,\mathrm{W}$
Load RMS voltage	$12.65\,\mathrm{V}$
Load peak voltage	$17.89\,\mathrm{V}$
Load peak-to-peak voltage	$35.78\,\mathrm{V}$
Load RMS current	$1.58\,\mathrm{A}$
Load peak current	$2.24\,\mathrm{A}$

2) Class AB Bias Adjustment: Bias sets the transistor operating conditions before an audio signal is applied. The design adopts a resting-current target of $I_D = 50\,\mathrm{mA}$ in each branch. This resting-current is established to keep on the transistors in a operating point to represent always all the audio signal variations.

This small current provides the intended overlap between the two transistors near the signal transition [**?**, sec. 7–2].

The $0.22\,\Omega$ source resistors help control branch current and provide a simple reference for adjustment. At the selected target, the voltage across each resistor is

$$V_{RS,Q} = I_Q R_S = (0.50)(0.22) = 11\,\mathrm{mV}. \tag{5}$$

The internal output is centered close to $0\,\mathrm{V}$ DC under this condition. To ensure the load (Speaker) only receives the audio signal.

The gate voltages are set by resistor dividers. For the resistor connected toward either supply rail, its effective value follows a voltage divider:

$$R_{\mathrm{bias}} = 100\,\mathrm{k\Omega}\left(\frac{24\,\mathrm{V}}{|V_G|} - 1\right) \tag{6}$$

Using nominal gate settings of $+3.6243\,\mathrm{V}$ and $-3.6123\,\mathrm{V}$ gives effective resistances of $562.2\,\mathrm{k\Omega}$ and $564.4\,\mathrm{k\Omega}$. Each branch therefore uses a commercial $560\,\mathrm{k\Omega}$ resistor in series with a $10\,\mathrm{k\Omega}$ trimmer. The effective trimmer settings are $2.2\,\mathrm{k\Omega}$ and $4.4\,\mathrm{k\Omega}$, corresponding to 22% and 44% of their nominal range.

The gate settings provide the initial bias references. The resting current is assessed through the source-resistor voltage drops. Table **??** keeps these design references separate from the recorded signal measurements.

3) Signal Coupling: The signal coupling is a factor to consider to keep isolated each stage of DC signals of polarization. Achieving that C1 and C2 pass the audio signal to the transistor gates while keeping their DC bias settings separate from the preceding stage.

This places the output coupling-network corner below $20\,\mathrm{Hz}$, helping preserve low-frequency audio.

4) Heat Dissipation: For the Class AB configuration when it always be bias also produces heat when no audio signal is present. At the selected resting-current target, each transistor dissipates approximately

$$P_{D,Q} \simeq V_{CC} I_Q = (24)(0.100) = 2.40\,\mathrm{W}. \tag{7}$$

During audio operation, dissipation is evaluated from the average power absorbed by each transistor, including the overlapping conduction of the Class AB stage.

A $10\,\mathrm{W}$ thermal design allowance per transistor provides margin above the recorded losses of $6.58\,\mathrm{W}$ and $6.63\,\mathrm{W}$. The heat-sink selection uses a thermal resistance of no more than $4\,^{\circ}\mathrm{C/W}$ per device. A lower thermal resistance allows heat to leave the transistor more easily. The selection also considers the device thermal data provided in the transistor datasheets [**?**], [**?**].

Table II
CALCULATED BIAS AND COUPLING DESIGN REFERENCES

Quantity	Design reference
Upper gate bias, initial	$+3.6243\,\mathrm{V}$
Lower gate bias, initial	$-3.6123\,\mathrm{V}$
Upper effective bias resistance	$562.2\,\mathrm{k\Omega}$
Lower effective bias resistance	$564.4\,\mathrm{k\Omega}$
R4 effective setting	$2.2\,\mathrm{k\Omega}$
R6 effective setting	$4.4\,\mathrm{k\Omega}$
Resting current, target per branch	$100\,\mathrm{mA}$
Source-resistor drop at target	$22\,\mathrm{mV}$ each
Internal output at rest, target	$0\,\mathrm{V}$
Resting transistor loss, estimate	$2.40\,\mathrm{W}$ each
Output coupling corner, estimate	$4.23\,\mathrm{Hz}$

IV. COMPONENT SELECTION CRITERIA

A. Preamplification Stage

B. Voltage Amplification Stage

C. Class AB Power Stage

The selected components retain the circuit topology and use commercial nominal values. Table **??** lists the main components and the ratings considered in their selection.

Table III
POWER-STAGE COMPONENTS AND SELECTION RATINGS

Reference	Value or model	Selection rating
U1	IRFZ44N	N-channel MOSFET
U2	IRF4905	P-channel MOSFET
R1, R5	$560\,\mathrm{k\Omega}$	1%, $0.25\,\mathrm{W}$
R2, R3	$100\,\mathrm{k\Omega}$	1%, $0.25\,\mathrm{W}$
R4, R6	$10\,\mathrm{k\Omega}$	Trimmers, $0.25\,\mathrm{W}$
RG1, RG2	$100\,\Omega$	$0.25\,\mathrm{W}$
RS1, RS2	$0.22\,\Omega$	$5\,\mathrm{W}$
C1, C2	$4.7\,\mu\mathrm{F}$	$50\,\mathrm{V}$, nonpolar
C3	$4700\,\mu\mathrm{F}$	$50\,\mathrm{V}$, nonpolar; suitable ripple rating
R_Z	$10\,\Omega$	$2\,\mathrm{W}$
C4	$100\,\mathrm{nF}$	$50\,\mathrm{V}$, film
SPK2	$8\,\Omega$	$30\,\mathrm{W}$ nominal rating

The MOSFET voltage ratings are considered together with the $\pm 24\,\mathrm{V}$ supplies and the expected signal excursions [**?**], [**?**]. The source-resistor power ratings provide margin for the load current, while the gate resistors and output RC network help control high-frequency behavior.

C3 carries the loudspeaker current, so its ripple-current rating accounts for the reported $1.69\,\mathrm{A}$ RMS operating value with margin. A nonpolar type accommodates the alternating voltage. The heat sinks are selected according to the dissipation allowance described above.

V. SIMULATION METHODOLOGY

A. Common Simulation Conditions

Multisim provides the circuit environment for the reported test. The power-stage input is a $20\,\mathrm{V}$ peak sinusoidal signal at $10\,\mathrm{kHz}$, with zero DC offset. The circuit uses nominal $\pm 24\,\mathrm{V}$ supplies and an $8\,\Omega$ load. R4 and R6 are set to 22% and 44%, respectively.

B. Preamplification Stage

C. Voltage Amplification Stage

D. Class AB Power Stage

1) DC Operating Point: The DC analysis examines the circuit with no audio input. It identifies the transistor bias and the internal output voltage. The design targets are $100\,\mathrm{mA}$ in each branch, $22\,\mathrm{mV}$ across each source resistor, and an internal output close to zero. These references define the intended Class AB resting condition.

2) Transient Response: The transient analysis follows the input and output signals over time. It provides the voltage, current, and average-power measurements reported in the results. The method compares settled signal cycles, allowing the output level and the transition between the two transistors to be examined under the same test conditions.

3) Frequency Response and Signal Quality: Frequency-response evaluation examines how the amplifier handles signals at different audio frequencies. Waveform and harmonic analyses assess changes in signal shape, including the transition between the positive and negative portions. These assessments complement the power measurements; the numerical readings presented here correspond to the documented $10\,\mathrm{kHz}$ test.

VI. SIMULATED VALUES AND RESULTS

A. Preamplification Stage

B. Voltage Amplification Stage

C. Class AB Power Stage

Tables **??** and **??** present the recorded values. RMS describes the effective signal level, peak-to-peak describes the full voltage or current swing, and mean describes the average DC component. The power readings are average values over the measurement interval.

Table IV
SIMULATED VOLTAGES AT $10\,\mathrm{kHz}$

Location	Peak-to-peak	RMS	Mean
IRFZ44N gate	$39.9\,\mathrm{V}$	$14.6\,\mathrm{V}$	$+3.62\,\mathrm{V}$
IRF4905 gate	$39.8\,\mathrm{V}$	$14.6\,\mathrm{V}$	$-3.62\,\mathrm{V}$
Output before C3	$38.2\,\mathrm{V}$	$13.5\,\mathrm{V}$	$+28.2\,\mathrm{mV}$
Loudspeaker load	$38.2\,\mathrm{V}$	$13.5\,\mathrm{V}$	$-87.2\,\mu\mathrm{V}$

The load voltage is $13.5\,\mathrm{V}$ RMS, with a mean close to zero after C3. This is consistent with the capacitor's role in passing the audio signal while blocking the steady DC component.

Table V
SIMULATED CURRENT AND AVERAGE POWER AT 10 kHz

Quantity	Simulated value
Load-current peak-to-peak	$4.78\,\mathrm{A}$
Load-current RMS	$1.69\,\mathrm{A}$
Load-current mean	$-235\,\mu\mathrm{A}$
Average loudspeaker-load power	$22.8\,\mathrm{W}$
IRFZ44N average dissipation	$6.58\,\mathrm{W}$
IRF4905 average dissipation	$6.63\,\mathrm{W}$
Signal frequency	$10.0\,\mathrm{kHz}$

The output-power reading exceeds the $20\,\mathrm{W}$ design reference. The transistor dissipation values describe heat produced within the output devices and are accounted for separately from the power received by the load.

VII. ANALYSIS AND INTEGRATION

A. Preamplification Stage

B. Voltage Amplification Stage

C. Class AB Power Stage

The recorded output voltage supports the average-power result. Applying (**??**) gives

$$P_L = \frac{(13.5)^2}{8} = 22.78\,\mathrm{W}, \tag{8}$$

which agrees with the $22.8\,\mathrm{W}$ simulation reading after rounding. The expected current is $13.5/8 = 1.69\,\mathrm{A}$ RMS, also consistent with the recorded value.

The $38.2\,\mathrm{V}$ peak-to-peak output corresponds to approximately $19.1\,\mathrm{V}$ peak. Compared with the specified $20\,\mathrm{V}$ peak input, the output amplitude is 95.5% of the input amplitude. This small reduction is consistent with the power stage's role of supplying current while preserving the signal polarity and most of its voltage amplitude.

At the reported frequency, the delivered power is 14% above the $20\,\mathrm{W}$ reference. The stage therefore meets the output-power objective under the documented test conditions. The two transistor losses are similar, and both remain below the $10\,\mathrm{W}$ per-device thermal design allowance.

For integration with the complete amplifier, the preceding stage provides the required input amplitude while driving the power-stage input. The bias references describe the resting condition, while the reported voltage, current, and power values describe operation with the audio test signal applied.

VIII. CONCLUSIONS AND NEXT STEPS

A. Preamplification Stage

B. Voltage Amplification Stage

C. Class AB Power Stage

The complementary IRFZ44N/IRF4905 stage retains the Class AB topology and uses commercial component values. Its bias design includes a $100\,\mathrm{mA}$ resting-current target, with trimmers for adjustment and source resistors that provide current references. Coupling capacitors pass the audio signal between the relevant circuit sections, and the thermal selection accounts for transistor power losses.

With a $20\,\mathrm{V}$ peak input at $10\,\mathrm{kHz}$, the recorded test delivers $22.8\,\mathrm{W}$ to an $8\,\Omega$ load, exceeding the $20\,\mathrm{W}$ design reference. The output reaches $13.5\,\mathrm{V}$ RMS and $1.69\,\mathrm{A}$ RMS. These results support the power-delivery objective at the stated test condition. Frequency-response, signal-quality, and connected-stage assessments complete the evaluation framework for the amplifier.

REFERENCES

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